

Wake-Up Stroke: Quick Guide

Wake-up strokes (WUS)—those discovered upon awakening without a known time of onset—account for up to 25% of ischemic strokes. Historically excluded from thrombolytic therapy, recent evidence supports treatment based on imaging. A 2024 individual patient-level meta-analysis (Kamogawa et al.) demonstrated that IV alteplase improved functional outcomes (mRS 0–1 at 90 days: 54.8% vs 45.5%, aOR 1.47; 95% CI, 1.01–2.16) in WUS without increased hemorrhagic risk.

For rural settings: assess as usual, using the time the patient was last seen normal (typically at bedtime). If MRI with DWI/FLAIR mismatch or CT perfusion shows viable tissue and suspected last known well within the time window, thrombolytic may be appropriate. When advanced imaging isn't available, early telestroke consultation and rapid transfer are essential. Facilities should pursue ASRH certification and leverage hub-and-spoke stroke networks to improve access.

Take-home: With appropriate imaging, WUS patients should not be excluded from thrombolysis. Use stroke tools such as telestroke, standardized protocols, and fast imaging to guide safe and effective treatment.

- Patient awakens with stroke symptoms → document LKW as bedtime.
- Rapid clinical assessment (NIHSS) + labs.
- Perform STAT MRI; if possible, advanced imaging to detect mismatch.
- Consult telestroke neurologist if available.
- If imaging favorable onset is within the time window to receive thrombolytic and no major contraindications, can consider.
- If imaging not available on-site, arrange timely transfer to stroke-capable center.
- Always follow institutional protocol—this is only a suggested approach.

Reference:

Kamogawa N, Miwa K, Toyoda K, et al. Thrombolysis for Wake-Up Stroke Versus Non-Wake-Up Unwitnessed Stroke: EOS Individual Patient Data Meta-Analysis. *Stroke*. 2024;55(4):895–904. doi:10.1161/STROKEAHA.123.043358.
